

[B19CS3101]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
III B. Tech I Semester (R19) Regular Examinations
DATA WAREHOUSING AND DATA MINING
Computer Science and Engineering
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks:75 M

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M											
UNIT - I																
1.	a).	Define data mining, data warehousing, KDD. List few applications of data mining.			CO1	K1	7									
	b).	Explain the different kinds of patterns that can be mined using data mining.			CO1	K2	8									
OR																
2.	a).	Given two objects represented by the tuples (22,1,42,10) and (20,0,36,8) a) Compute Euclidean distance between the objects b) Compute Manhattan distance between objects d) Compute the Supremum distance between the two objects			CO1	K3	7									
	b).	Discuss in detail about major issues inDataMining.			CO1	K2	8									
UNIT - II																
3.	a).	Summarize the different data preprocessing techniques.			CO2	K2	8									
	b).	Develop a linear regression model to the following dataset			CO2	K3	7									
		Car age(In years)	4	4				5	5	7	7	8	9	10	11	12
		Price (in dollars)	6300	5800				5700	4500	4200	4100	3100	2100	2500	2200	
OR																
4.	a).	Compare OLAP and OLTP systems			CO2	K2	7									
	b).	Illustrate the multitier datawarehouse architecture			CO2	K2	8									

UNIT - III																												
5.	a).	Define frequent itemset, closed itemset and maximal itemset. A database has five transactions. Let min_sup D 60% and min_conf 80%. Apply Apriori algorithm to find all frequent itemsets.	CO3	K3	8																							
		<table><tr><td>TID</td><td>Items_bought</td></tr><tr><td>T100</td><td>{M,O,N,K,E,Y}</td></tr><tr><td>T200</td><td>{D,O,N,K,E,Y}</td></tr><tr><td>T300</td><td>{M,A,K,E}</td></tr><tr><td>T400</td><td>{M,U,C,K,Y}</td></tr><tr><td>T500</td><td>{C,O,O,K,I,E}</td></tr></table>	TID	Items_bought	T100	{M,O,N,K,E,Y}	T200	{D,O,N,K,E,Y}	T300	{M,A,K,E}	T400	{M,U,C,K,Y}	T500	{C,O,O,K,I,E}														
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T400	{M,U,C,K,Y}																											
T500	{C,O,O,K,I,E}																											
	b).	Demonstrate FP Growth algorithmwithan example.	CO3	K2	7																							
OR																												
6.	a).	Outline the different pattern evaluation methods.	CO3	K2	7																							
	b).	Explain about mining multilevel association.	CO3	K2	8																							
UNIT - IV																												
7.	a).	Demonstrate the construction of a Decision tree with an example.	CO4	K2	8																							
	b).	For the following confusion matrix generated by a classifier, Calculate the following parameters i) The accuracy of classifier ii) Precision and Recall of each class	CO4	K3	7																							
		<table><tr><td></td><td colspan="4">Predicted Class</td></tr><tr><td rowspan="4">Actual Class</td><td></td><td>A</td><td>B</td><td>C</td></tr><tr><td>A</td><td>14</td><td>3</td><td>3</td></tr><tr><td>B</td><td>9</td><td>15</td><td>6</td></tr><tr><td>C</td><td>1</td><td>2</td><td>47</td></tr></table>		Predicted Class				Actual Class		A	B	C	A	14	3	3	B	9	15	6	C	1	2	47				
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	C	1	2	47																								
OR																												

8.	a).	Apply Naïve Bayes algorithm (or Decision Tree Induction) on the following dataset and classify the following tuple (CO4, K3) <i>(Outlook=Sunny, Temperature=Cool, Humidity=High, Wind=Strong)</i> <table border="1"> <thead> <tr> <th></th><th colspan="4">Predictors</th><th>Response</th></tr> <tr> <th></th><th>Outlook</th><th>Temperature</th><th>Humidity</th><th>Wind</th><th>Class</th></tr> </thead> <tbody> <tr> <td></td><td></td><td></td><td></td><td></td><td>Play=Yes</td></tr> <tr> <td></td><td></td><td></td><td></td><td></td><td>Play=No</td></tr> <tr> <td>Day1</td><td>Sunny</td><td>Hot</td><td>High</td><td>Weak</td><td>No</td></tr> <tr> <td>Day2</td><td>Sunny</td><td>Hot</td><td>High</td><td>Strong</td><td>No</td></tr> <tr> <td>Day3</td><td>Overcast</td><td>Hot</td><td>High</td><td>Weak</td><td>Yes</td></tr> <tr> <td>Day4</td><td>Rain</td><td>Mild</td><td>High</td><td>Weak</td><td>Yes</td></tr> <tr> <td>Day5</td><td>Rain</td><td>Cool</td><td>Normal</td><td>Weak</td><td>Yes</td></tr> <tr> <td>Day6</td><td>Rain</td><td>Cool</td><td>Normal</td><td>Strong</td><td>No</td></tr> <tr> <td>Day7</td><td>Overcast</td><td>Cool</td><td>Normal</td><td>Strong</td><td>Yes</td></tr> <tr> <td>Day8</td><td>Sunny</td><td>Mild</td><td>High</td><td>Weak</td><td>No</td></tr> <tr> <td>Day9</td><td>Sunny</td><td>Cool</td><td>Normal</td><td>Weak</td><td>Yes</td></tr> <tr> <td>Day10</td><td>Rain</td><td>Mild</td><td>Normal</td><td>Weak</td><td>Yes</td></tr> <tr> <td>Day11</td><td>Sunny</td><td>Mild</td><td>Normal</td><td>Strong</td><td>Yes</td></tr> <tr> <td>Day12</td><td>Overcast</td><td>Mild</td><td>High</td><td>Strong</td><td>Yes</td></tr> <tr> <td>Day13</td><td>Overcast</td><td>Hot</td><td>Normal</td><td>Weak</td><td>Yes</td></tr> <tr> <td>Day14</td><td>Rain</td><td>Mild</td><td>High</td><td>Strong</td><td>No</td></tr> </tbody> </table>		Predictors				Response		Outlook	Temperature	Humidity	Wind	Class						Play=Yes						Play=No	Day1	Sunny	Hot	High	Weak	No	Day2	Sunny	Hot	High	Strong	No	Day3	Overcast	Hot	High	Weak	Yes	Day4	Rain	Mild	High	Weak	Yes	Day5	Rain	Cool	Normal	Weak	Yes	Day6	Rain	Cool	Normal	Strong	No	Day7	Overcast	Cool	Normal	Strong	Yes	Day8	Sunny	Mild	High	Weak	No	Day9	Sunny	Cool	Normal	Weak	Yes	Day10	Rain	Mild	Normal	Weak	Yes	Day11	Sunny	Mild	Normal	Strong	Yes	Day12	Overcast	Mild	High	Strong	Yes	Day13	Overcast	Hot	Normal	Weak	Yes	Day14	Rain	Mild	High	Strong	No	CO4	K3	7
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	b).	Discuss about Back propagation algorithm for neural network-based classification of data.	CO4	K2	8																																																																																																												
UNIT - V																																																																																																																	
9.	a).	Explain about different types of data in cluster analysis	CO5	K2	8																																																																																																												
	b).	Consider that the data mining task is to cluster the following seven points P1, P2, P3, P4, P5, P6, P7 into two clusters. P1 (1,1), P2 (2,2), P3 (3,4), P4 (5,7), P5 (3,5), P6 (4,5) and P7(4,6). The distance function is Euclidean distance. Apply K-means algorithm with two iterations to form two clusters by taking the initial cluster centres as points P1 and P4.	CO5	K3	7																																																																																																												
OR																																																																																																																	
10.	a).	Demonstrate DBScan algorithm with the help of an example	CO5	K2	8																																																																																																												
	b).	Compare partitional and hierarchical clustering algorithms.	CO5	K4	7																																																																																																												

[B19CS3102]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
III B. Tech I Semester (R19) Regular Examinations
COMPUTERNETWORKS
Computer Science and Engineering
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks:75 M

Answer **ONE Question** from **EACH UNIT**
All questions carry equal marks

			CO	KL	M
		UNIT-I			
1	a.	Define Switching. Explain about the Message switching.	1	2	8
	b.	Discuss about the OSI Reference model with neat sketch.	1	2	7
		OR			
2	a.	Differentiate between the Packet Switching and Circuit Switching	1	3	8
	b.	Explain any two LAN Topologies with a neat diagram	1	2	7
		UNIT-II			
3	a.	Identify any two Line Encoding Techniques.	2	2	8
	b.	Demonstrate about the FDM and TDM multiplexing with neat diagrams.	2	3	7
		OR			
4	a.	Illustrate about the selective repeat protocol mechanism.	2	3	8
	b.	Define aloha and explain about the Slotted aloha	2	3	7
		UNIT-III			
5	a.	Solve the following. A bit stream 1101011011 is transmitted using the standard CRC method. The generator polynomial is x^4+x+1 . What is the actual bit string transmitted?	3	3	8
	b.	Explain about the wireless communication standard Wi-Fi with neat diagram.	3	2	7
		OR			
6	a.	Compare between the Fast Ethernet and Ethernet in details.	3	3	8
	b.	Explain about the token ring mechanism in detail.	3	2	7
		UNIT-IV			
7	a.	Illustrate about the IPv4 header with neat diagram.	4	3	8
	b.	Demonstrate about the Distance Vector Routing Algorithm with example.	4	3	7
		OR			

8	a.	Discuss about the DHCP protocol with neat diagram.	4	2	8
	b.	Define subnetting and explain with an example about subnetting.	4	3	7
		UNIT-V			
9	a.	Define protocol. Explain about the TCP protocol with neat diagram.	5	3	8
	b.	Define File Transfer. Write short notes about FTP protocol	5	3	7
		OR			
10	a.	Describe about the congestion control algorithms	5	3	8
	b.	Write short notes about DNS	5	3	4
	C.	Define HTTP. List out some of the basic operations of HTTP.	5	2	3

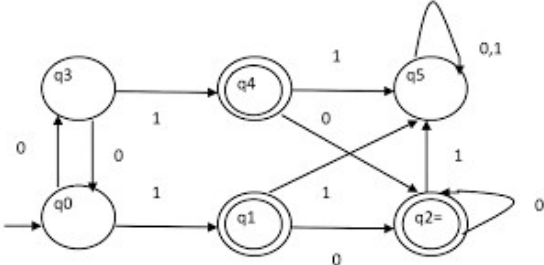
[B19 CS 3103]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
III B. Tech I Semester (R19) Regular Examinations
FORMAL LANGUAGES AND AUTOMATA THEORY
Computer Science and Engineering
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks:75 M

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
UNIT - I					
1.	a).	Construct DFA for the following languages i) $L = \{w/w \text{ begins with } 1 \text{ and ends with } 00, w \text{ in } \{0,1\}^*\}$ ii) $L = \{w/w \text{ contains Even number of zeros and Even number of ones, } w \text{ in } \{0,1\}^*\}$	CO1	K3	8
	b).	Differentiate Mealy and Moore Machines, Give an example for Mealy machine and Moore machine.	CO1	K2	7
OR					
2.	a).	Construct NFA for accepting the strings $\{ab, ba\}$ and then convert it to DFA.	CO1	K3	8
	b).	Explain about minimization algorithm. And apply minimization algorithm to reduce the number of states of following DFA 	CO1	K2	7
UNIT - II					
3.	a).	Define Regular expression. Construct NFA with ϵ moves equivalent to the Regular Expression $(ab + aab)^*$	CO2	K3	7
	b).	Identify the Regular expression and construct the equivalent Finite automata for the following language descriptions : i. set of all strings beginning with 01 and ending with 10 ii. set of all strings having three consecutive zeros or three consecutive ones	CO2	K3	8
OR					
4.	a).	State Pumping lemma on Regular sets; list the applications of pumping lemma.	CO2	K1	7

	b).	Apply pumping lemma to show the set of all even length palindrome strings is not regular.	CO2	K3	8
UNIT - III					
5.	a).	Construct Context free grammar for $L = \{ WCW^R / W \text{ in } (0+1)^* \}$	CO3	K3	8
	b).	Define CNF and convert the following CFG to CNF $S \rightarrow aSa / bSb / a / b$	CO3	K2	7
OR					
6.	a).	Construct Context free grammar for generating all palindrome strings over $(0,1)$	CO3	K3	8
	b).	Define GNF and Convert the following CFG to GNF $S \rightarrow AA \mid a \quad A \rightarrow SS \mid b$	CO3	K2	7
UNIT - IV					
7.	a).	Define Pushdown Automata? Explain the acceptance of PDA by empty stack using an example.	CO4	K2	7
	b).	Construct PDA for recognizing the Context free language $L = \{ a^n c b^n / n \geq 1 \}$	CO4	K3	8
OR					
8.	a).	What is ID of PDA. Explain the acceptance of PDA by final state	CO4	K2	7
	b).	Construct a PDA to accept language of odd length palindrome strings	CO4	K3	8
UNIT - V					
9.	a).	Define the Turing Machine and Explain different types of TM?	CO5	K2	7
	b).	Construct a TM for recognizing the language $L = \{ WW^R / W \text{ in } (a,b)^* \}$	CO5	K3	8
OR					
10.	a).	What is PCP? Give an example	CO5	K2	7
	b).	Explain about P and NP classes	CO5	K2	8

[B19CS3104]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
III B. Tech I Semester (R19) Regular Examinations
WEB TECHNOLOGIES
Computer Science and Engineering
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks:75

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

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			CO	KL	M
UNIT – I					
1.	a).	Explain the concept of frameset and frames in building Web Pages	CO1	K2	7
	b).	Explain various form elements with an example	CO1	K2	8
OR					
2.	a).	Explain different types of lists in HTML with an example	CO1	K2	7
	b).	Write an HTML script to manage personal details of a student like name, class, qualifications, photo, address etc., using tables and other suitable tags.	CO1	K2	8
UNIT – II					
3.	a).	Explain various types of Style sheets with example	CO2	K2	7
	b).	Write a java script to print Armstrong Numbers between 1 to 500.	CO2	K2	8
OR					
4.	a).	Explain different java script objects.	CO2	K2	7
	b).	Explain various types of selectors in CSS with suitable examples	CO2	K2	8
UNIT – III					
5.	a).	Explain about Ajax Mechanism with an Example	CO3	K2	7
	b).	Explain the differences between DOM & SAX	CO3	K2	8
OR					
6.	a).	Explain about XML schema with an example	CO3	K2	8
	b).	Explain about integration of Ajax and PHP with an example	CO3	K2	7
UNIT – IV					
7.	a).	Explain about the types of Arrays in PHP with an example.	CO4	K2	7
	b).	Explain any 5 String Operations in PHP with Suitable Examples	CO4	K2	8
OR					
8.	a).	Explain various types of control statements in PHP	CO4	K2	8
	b).	Write a PHP Script to find the Fibonacci sequence upto a given number	CO4	K2	7
UNIT – V					
9.	a).	Explain how a Session is created and Destroyed in PHP	CO5	K2	7
	b).	Develop a PHP program to fetch data from the MYSQL database.	CO5	K3	8
OR					
10.	a).	Develop a PHP Program to Insert data into a MYSQL database	CO5	K3	8
	b).	Explain about Cookies in PHP with an example.	CO5	K2	7

[B19CS3105]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
III B. Tech I Semester (R19) Regular Examinations
COMPUTER GRAPHICS
Computer Science and Engineering
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks:75 M

Answer **ONE Question** from **EACH UNIT**
 All questions carry equal marks

			CO	KL	M
UNIT – I					
1.	a).	Explain applications areas of Computer Graphics?	CO1	K2	8
	b).	Describe the Raster-Scan system with illustrations.	CO1	K2	7
OR					
2.	a).	Explain about CRT monitors	CO1	K2	8
	b).	Describe about Logical classification of Input Devices	CO1	K2	7
UNIT – II					
3.	a).	Illustrate DDA Line drawing algorithm using an example.	CO2	K3	8
	b).	Scan convert a circle with radius 5 units and centre point (5,5)using Mid-Point Circle Algorithm.	CO2	K3	7
OR					
4.	a).	Scan convert a line starting at (0,0) and ending at (10,6) using Brenham's Line drawing algorithm.	CO2	K3	7
	b).	Illustrate Flood fill algorithm.	CO2	K3	8
UNIT – III					
5.	a).	Demonstrate Two-Dimensional translation, rotation and scaling.	CO3	K3	8
	b).	Obtain the coordinates of a square with (-2,-2), (2,-2), (2,2)and (-2,2) as the corner points after rotating it by 90 degrees clockwise about the point (2,2).	CO3	K3	7
OR					
6.	a).	Derive transformation matrix for 2D fixed point scaling.	CO3	K3	7
	b).	Derive transformation matrix for 2D pivot point rotation.	CO3	K3	8
UNIT – IV					
7.	a).	Derive the transformation matrix for Window-to-Viewport coordinate transformation.	CO5	K3	7
	b).	Demonstrate Cohen-Sutherland line Clipping Algorithm with diagrams.	CO5	K3	8
OR					
8.	a).	Illustrate Liang-Barsky line clipping algorithm.	CO5	K3	8
	b).	Apply Sutherland-Hodgeman algorithm for polygon Clipping.	CO5	K3	7
UNIT – V					
9.	a).	Construct Three-Dimensional translation and scaling transformation matrices.	CO4	K3	8

	b).	Illustrate natural cubic spline	C05	K3	7
OR					
10.	a).	Derive transformation matrix for 3D Rotation about an arbitrary line.	C04	K3	8
	b).	Illustrate the properties of Bezier curves.	C05	K3	7

[B19CS3106]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
III B. Tech I Semester (R19) Regular Examinations
PRINCIPLES OF PROGRAMMING LANGUAGES
Computer Science and Engineering
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
UNIT - I					
1.	a).	Discuss in detail about attribute grammars.	CO1	K2	7
	b).	Explain different phases of Compilation.	CO1	K3	8
OR					
2.	a).	Discuss the criteria for evaluating a programming language.	CO1	K2	8
	b).	Explain about Lexical Analysis.	CO1	K3	7
UNIT - II					
3.	a).	Explain in detail arrays, indices, subscript bindings, and array categories	CO2	K2	8
	b).	Explain in detail about overloaded operators.	CO2	K2	7
OR					
4.	a).	Explain Scope and lifetime of variables.	CO2	K2	7
	b).	Explain about conditional statements and implementations with examples.	CO2	K2	8
UNIT - III					
5.	a).	Explain the design issues for functions.	CO3	K3	7
	b).	Explain how subprogram names are passed as parameters. Write the semantics of call and return of a subprogram.	CO3	K2	8
OR					
6.	a).	Discuss how generic methods are implemented with suitable examples.	CO3	K2	7

	b).	Explain the importance of Dynamic Scoping with an example.	C03	K2	8
UNIT – IV					
7.	a).	Discuss design issues for various OOP languages.	C04	K2	8
	b).	Explain how events are handled in various OOP languages	C04	K3	7
OR					
8.	a).	Explain how cooperation synchronization and competition synchronization are implemented using monitors.	C04	K3	8
	b).	Discuss about Exception Handling in C++.	C04	K2	7
UNIT – V					
9.	a).	Explain about fundamentals of Functional Programming Language.	C05	K2	7
	b).	Differentiate ML and other Fundamental Programming Languages.	C05	K4	8
OR					
10.	a).	Explain the applications of Logic Programming.	C05	K3	7
	b).	Explain the interfacing process of PROLOG.	C05	K2	8

[B19CS3107]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
III B. Tech I Semester (R19) Regular Examinations
SCRIPTING LANGUAGES
Computer Science and Engineering
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**
 All questions carry equal marks

			CO	KL	M
UNIT - I					
1.	a).	Explain the characteristics of PERL with examples	CO1	K2	7
	b).	Explain about HTML tags emulation, arrays and hashes used in PERL	CO1	K3	8
OR					
2.	a).		CO1	K2	8
	b).	Explain about subroutines in PERL with example	CO1	K3	7
UNIT - II					
3.	a).	Explain about packages in PERL with example	CO2	K2	8
	b).	Explain about security issues in PERL	CO2	K3	7
OR					
4.	a).	Write a PHP program to add array elements taken as input	CO2	K2	7
	b).	Discuss about Associative Array in PHP with example	CO2	K2	8
UNIT - III					
5.	a).	Write a PHP Script to retrieve the data from database	CO3	K2	8
	b).	Explain about PHP Encryption Functions with example	CO3	K3	7
OR					
6.	a).	Write a PHP Script to insert data into the database	CO3	K2	8
	b).	Explain about the M Crypt Package in PHP	CO3	K3	7
UNIT - IV					
7.	a).	Discuss about variables and data in TCL	CO4	K2	8
	b).	Discuss about security issues in TCL	CO4	K2	7
OR					
8.	a).	Discuss about exec and uplevel commands in TCL	CO4	K2	8
	b).	Explain about Events and Binding in TCL	CO4	K3	7
UNIT - V					
9.	a).	Explain about Control structures in Python with example	CO5	K2	8
	b).	Explain about Built-in functions and methods in Python	CO5	K2	7
OR					
10.	a).	Explain about web application Frame work in Python	CO5	K2	8
	b).	Explain about Python web systems	CO5	K3	7

[B19CS3108]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
III B. Tech I Semester (R19) Regular Examinations
ADVANCED COMPUTER ARCHITECTURE
Computer Science and Engineering
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
		UNIT-I			
1	a.	Discuss different Classes of Computing Applications and their Characteristics	1	2	8
	b.	Explain the Eight Great Ideas in Computer Architecture.	1	3	7
		OR			
2	a.	Explain different technologies for Building Processors and Memory	1	3	7
	b.	Discuss about the Power wall	1	2	8
		UNIT-II			
3	a.	Discuss about different signed and unsigned numbers with an example	2	2	8
	b.	Explain the different types of MIPS addressing modes with an example	2	3	7
		OR			
4	a.	Illustrate the steps to transform a c file on a disk into a program running on a computer.	2	2	8
	b.	Explain different instructions and registers in ARM 32bit Architecture.	2	3	7
		UNIT-III			
5	a.	Explain Sequential Version of the Multiplication Algorithm and Hardware	3	3	7
	b.	Illustrate Booth Algorithm with an example.	3	2	8
		OR			
6	a.	Explain briefly about Binary Addition and Subtraction in MIPS architecture	3	3	8
	b.	List and explain various floating-Point Instructions in MIPS	3	2	7
		UNIT-IV			
7	a.	Describe the data path and control considerations for pipelining.	4	2	7
	b.	Explain the methods for dealing with the instruction hazards.	4	3	8
		OR			
8	a.	Briefly explain about Creating a Single Data path with neat diagram	4	2	8
	b.	Describe the techniques for handling data and instruction hazards in pipelining	4	2	7
		UNIT- V			
9	a.	Describe the performance consideration of cache memory	5	2	8
	b.	Explain the concept of virtual memory with any one virtual memory management technique.	5	3	7
		OR			
10	a.	Explain the concept of memory hierarchy	5	2	7

	b.	Explain how the virtual address is converted into real address in a paged virtual memory system	5	3	8
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[B19CS3109]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
III B. Tech I Semester (R19) Regular Examinations
SOFTWARE TESTING METHODOLOGIES
Computer Science and Engineering
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
UNIT - I					
1.	a).	Explain about Software Testing Life Cycle?	CO1	K3	7
	b).	Differentiate Verification and Validation Activities in Testing Methodology?	CO1	K4	8
OR					
2.	a).	Discuss in detail about evolution of software testing .	CO1	K2	8
	b).	Discuss about Myths related software testing and its facts	CO1	K2	7
UNIT - II					
3.	a).	Classify various types of errors detected by black box testing?	CO2	K2	8
	b).	Explain about Decision table based testing	CO2	K3	7
OR					
4.	a).	What is the need of white box testing? Discuss briefly	CO2	K2	7
	b).	Explain about Basis Path Testing?	CO2	K3	8
UNIT - III					
5.	a).	Explain about the Function Testing with an example?	CO3	K2	8
	b).	Explain about Unit Testing and Integration Testing?	CO3	K3	7
OR					
6.	a).	Differentiate Progressives and regressive testing ?	CO3	K4	8
	b).	Classify the types of Regression Testing?	CO3	K3	7
UNIT - IV					
7.	a).	Explain the nature of Test Suits and Test Benefits?	CO4	K3	8
	b).	Classify the Prioritization Techniques ?	CO4	K2	7
OR					
8.	a).	Explain about the Software Quality Metrics?	CO4	K2	8
	b).	Explain about the Process of SQA Models Debugging?	CO4	K3	7
UNIT - V					
9.	a).	Explain about the Testing tools and Guidelines for Automated Testing?	CO5	K3	8
	b).	Explain about Jmeter and Junit?	CO5	K2	7
OR					
10.	a).	Classify Challenges in testing for web based software?	CO5	K2	8
	b).	Discuss in detail about Testing of mobile systems?	CO5	K3	7

[B19HS3201]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
III B. Tech II Semester (R19) Regular Examinations
MANAGERIAL ECONOMICS AND FINANCIAL ACCOUNTANCY
Computer Science and Engineering
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**
All questions carry equal marks

S.No.	Question / Problem	CO	KL	M
	UNIT-I			
1.	Define Managerial Economics and Explain its nature and scope	CO1	K2	15
	OR			
2.	What do you mean by Elasticity of demand? Explain in detail about degrees of Price elasticity of Demand?	CO1	K2	15
	UNIT-II			
3.	Define Cost & classify the Elements of Cost?	CO2	K2	15
	OR			
4.	How do you calculate BEP? What are its Assumptions and Applications?	CO2	K3	15
	UNIT-III			
5.	What are Market Structures and explain the features of Perfect Competition?	CO3	K2	15
	OR			
6.	Why is pricing significant in the context of business? Describe any four pricing practices?	CO3	K2	15
	UNIT-IV			
7.	Describe about the Importance of Accounting and types of accounts	CO4	K2	15
	OR			
8.	From the following Trail Balance of Suresh as at December 31, 2013, prepare Trading, Profit and Loss Account for the year ended December 31, 2013 and a Balance Sheet as on that date:	CO4	K3	15

	Dr. (Rs.) Cr. (Rs.)				
	Purchases of materials	32,000			
	Productive wages	13,000			
	Sales	60,000			
	Salaries	4,000			
	Travelling expenses	1,000			
	Carriage inwards	550			
	Insurance	300			
	Commission	650			
	Rent and rates	1,000			
	Cash in hand	350			
	Cash at bank	5,550			
	Repairs	600			
	Sundry expenses	110			
	Mortgage	6,100			
	Buildings	8,000			
	Machinery	3,000			
	Furniture	1,000			
	Stock on hand (1.1.2013)	11,500			
	Capital	21,310			
	Sundry debtors	9,000			
	Sundry creditors	4,200			
		91,610	91,610		

	Adjust the following: (a) Prepaid rent Rs. 100 (b) Depreciate the following: • Buildings @ 10per cent per annum • Machinery @ 20 per cent per annum • Furniture @ 15 per cent per annum (c) Provide for bad debts Rs. 100 (d) Outstanding insurance Rs. 50 (e) Closing stock Rs. 12,000			
	UNIT-V			
9.	Explain about capital and the sources available for raising finance	CO5	K2	15
	OR			
10.	Explain about the concept and causes of depreciation. Evaluate the straight-line method and diminishing balance methods.	CO5	K2	15

[B19 CS 3201]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
III B. Tech II Semester (R19) Regular Examinations
COMPILER DESIGN
Computer Science and Engineering
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks:75 M

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
UNIT - I					
1.	a).	Explain about the structure of a compiler with neat diagram.	CO1	K2	8
	b).	Explain the role of Lexical analysis.	CO1	K2	7
OR					
2.	a).	Show the output of different phases of compiler for the statement POS: = INITIAL + RATE * 60.	CO1	K3	8
	b).	Explain about Recognition of tokens.	CO1	K2	7
UNIT - II					
3.	a).	Explain the role of parser	CO2	K2	7
	b).	Explain how the Shift-Reduce parser operates by taking an example.	CO2	K2	8
OR					
4.	a).	List the differences between Top-down and Bottom-up parsers and eliminate the left recursion in CFG with an example.	CO2	K3	7
	b).	Explain about Predictive parser.	CO2	K2	8
UNIT - III					
5.	a).	What is LR parser and construct SLR parsing table using a CFG	CO3	K3	8
	b).	Explain about Dependency graph	CO3	K2	7
OR					
6.	a).	Explain CLR parser by taking an example.	CO3	K3	8

	b).	Explain about evaluation of SDD at nodes of a parse tree by taking an example	CO3	K2	7
UNIT – IV					
7.	a).	Explain about different types of Three-Address code statements	CO4	K2	8
	b).	Explain about the principle sources of optimization	CO4	K2	7
OR					
8.	a).	Explain about DAG and construct DAG for $a + a*(b-c) + (b-c)*d$	CO4	K3	8
	b).	Explain about optimization of basic blocks.	CO4	K2	7
UNIT – V					
9.	a).	List the techniques used under peephole optimization and explain each technique.	CO5	K2	8
	b).	Explain about Heap Management.	CO5	K2	7
OR					
10.	a).	Explain about simple code generator.	CO5	K2	8
	b).	Explain about storage organization	CO5	K2	7

[B19 CS 3202]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
III B. Tech II Semester (R19) Regular Examinations
UML & DESIGN PATTERNS
Computer Science and Engineering
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks:75

Answer **ONE Question** from **EACH UNIT**
 All questions carry equal marks

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			CO	KL	M
UNIT - I					
1.	a).	Explain the Basic Principles of Modelling?	CO1	K2	7
	b).	Explain the phases of a Unified Process with a neat sketch?	CO1	K2	8
OR					
2.	a).	Explain the importance of Modelling?	CO1	K2	7
	b).	Describe Basic Modelling Techniques for designing the system?	CO1	K2	8
UNIT - II					
3.	a).	Explain different elements in Use case diagram with an example?	CO2	K2	7
	b).	Draw State chart diagram for railway management system	CO2	K2	8
OR					
4.	a).	Explain the uses and basic elements in Activity diagram with examples	CO2	K2	7
	b).	Draw a sequence diagram for performing any ATM transaction	CO2	K2	8
UNIT - III					
5.	a).	Explain about Component diagram and Deployment diagram with an example?	CO3	K2	7
	b).	In a factory, there is a automated furnace system. The furnace can be controlled only through temperature in fahrenheit format. But Furnace Regulatory systems gets temperature in centigrade format. Draw the UML diagram considering appropriate Design Pattern.	CO3	K2	8

OR					
6.	a).	Explain the MVC Pattern? List and briefly explain the benefits of MVC Pattern	CO3	K2	8
	b).	Classify and briefly explain the design patterns.	CO3	K2	7
UNIT – IV					
7.	a).	In which context Singleton Pattern is Applicable Explain it	CO4	K2	7
	b).	Explain Applicability of Façade Pattern with proper Example	CO4	K2	8
OR					
8.	a).	In which context Builder Pattern is Applicable Explain it?	CO4	K2	7
	b).	Explain Applicability of Bridge Pattern with proper Example	CO4	K2	8
UNIT – V					
9.	a).	Explain about Interpreter with an Example?	CO5	K2	7
	b).	Explain about the chain of Responsibility?	CO5	K2	8
OR					
10.	a).	Explain about the Template Method with an Example?	CO5	K2	8
	b).	Classify the Collaborations and Consequences with an Example?	CO5	K2	7

[B19 CS 3203]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
III B. Tech II Semester (R19) Regular Examinations
ARTIFICIAL INTELLIGENCE
Computer Science and Engineering
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks:75 M

Answer **ONE Question** from **EACH UNIT**
 All questions carry equal marks

			CO	KL	M
UNIT - I					
1.	a).	Explain about different problem characteristics of AI problem.	CO1	K2	8
	b).	Differentiate BFS and DFS	CO1	K2	7
OR					
2.	a).	Explain about water jug problem and suggest a suitable solution to water jug problem.	CO1	K2	8
	b).	Apply BFS algorithm to solve water jug problem	CO1	K3	7
UNIT - II					
3.	a).	Explain A* algorithm with an example.	CO2	K2	8
	b).	Apply nearest neighbor heuristic algorithm to solve TSP.	CO2	K3	7
OR					
4.	a).	Explain about Means ends analysis.	CO2	K2	8
	b).	Apply simple hill climbing to solve 8-puzzle problem.	CO2	K3	7
UNIT - III					
5.	a).	Apply unification algorithm to the following : a. Like(john, x) Hate (john, x) b. Like (Marcus, Caesar, john) and Like(x, y) c. Like (john, kate) and Like (x, kate)	CO3	K3	8
	b).	Explain Truth maintenance system in detail.	CO3	K2	7
OR					

6.	a).	- Apply the resolution algorithm to Prove that: - John likes peanuts. From the following facts: - John likes all kind of food. - Apple and vegetable are food. - Anything anyone eats and not killed is food. - Anil eats peanuts and still alive. - Harry eats everything that Anil eats.	C03	K3	8
	b).	Explain about Clause conversion algorithm.	C03	K2	7
UNIT - IV					
7.	a).	Differentiate Procedural knowledge and Declarative Knowledge.	C04	K2	7
	b).	Represent the following facts using Partitioned Semantic Nets: - The dog bite the mail carrier - Every batter hit every bowler	C04	K3	8
OR					
8.	a).	Differentiate Forward and Backward Reasoning.	C04	K3	7
	b).	Represent the following facts in Conceptual Dependency : - John gave the AI book to marry. - John punched marry.	C04	K3	8
UNIT - V					
9.	a).	Explain about goal stack planning.	C05	K3	7
	b).	Explain different types of expert systems.	C05	K2	8
OR					
10.	a).	Explain various steps in the natural language processing.	C05	K2	7
	b).	Explain the architecture of Expert system.	C05	K3	8